

Multi-Physics Modeling of Mass and Thermal Transport Dynamics in Li-Ion Batteries vs. Semiconductor Silicon Wafers using MATLAB

Abstract:

This project delivers a MATLAB-based multi-physics simulation evaluating the mass and thermal transport kinetics of Li-ion battery cathodes versus silicon semiconductor lattices. By numerically solving transient Fick's and Fourier's transport equations, the model quantifies the stark scaling divergence between fast interstitial lithium intercalation (**10 – 120 seconds**) and slow substitutional boron dopant diffusion (**0.5 – 4 hours**). Furthermore, it maps the thermal contrast between silicon's ultra-fast dissipation (**0.1 – 2 milliseconds**) and the severe heat retention of ceramic battery oxide matrices. These quantitative profiles provide essential engineering baselines for designing active battery cooling systems and microelectronic thermal budgets.

Introduction:

Thermal and mass transport limitations dictate the performance boundaries of both high-capacity battery packs and dense microchip architectures. If a battery accumulates heat faster than it can dissipate it, it risks thermal runaway; if a semiconductor logic gate overheats, it suffers structural damage or performance throttling.

This project uses **MATLAB** to simulate these exact physical limits across a micro-scale zone (**0 to 5 μm**). The objective is twofold:

- ❑ **Quantify Mass Transport:** Contrast the efficiency of interstitial lithium-ion intercalation against high-barrier substitutional boron dopant diffusion.
- ❑ **Quantify Thermal Transport:** Analyze the heat retention of low-conductivity ceramic oxides (NMC cathodes) against the rapid dissipation of crystalline semiconductors (Silicon).

Methodology:

The spatial-temporal distribution of mass and energy is modeled using non-steady-state partial differential equations (PDEs) under a 1D semi-infinite medium framework with a constant surface source.

(i). Governing Laws:

Mass Diffusion (Fick's Second Law): Predicts the change in concentration gradient over time under non-steady-state diffusion conditions:

$$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2}$$

Thermal Conduction (Fourier's Second Law): Describes how temperature changes over time within a material as heat flows via conduction:

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

Where T represents the temperature rise (**T**) and α is the thermal diffusivity. Diffusivity is a function of thermal conductivity (**k**), density (ρ), and specific heat capacity (C_p):

$$\alpha = \frac{k}{\rho C_p}$$

(ii). Analytical Solution Framework: For a semi-infinite medium with a constant source at the surface ($x = 0$) and an initial bulk concentration value of zero, the transport equations resolve analytically via the Complementary Error Function (**erfc**):

$$C = C_o \cdot \operatorname{erfc}\left(\frac{x}{2\sqrt{Dt}}\right)$$

$$\Delta T = \Delta T_o \cdot \operatorname{erfc}\left(\frac{x}{2\sqrt{\alpha t}}\right)$$

(iii). Modeling Assumptions and limitations: To maintain computational efficiency while capturing core transport physics, the model applies the following engineering assumptions:

- 1. **1D Unidirectional Transport:** Transport across the planar surface area (y, z) is assumed uniform, reducing the PDE to a 1D spatial variable (x).
- 2. **Isotropic & Constant Properties:** Diffusivity coefficients (D, α) are treated as constant across the localized temperature step ($\Delta T = 50^{\circ}\text{C}$)
- 3. **Negligible Side Kinetics:** Volumetric heat generation (Joule heating inside the battery) and structural phase changes are neglected to isolate pure conduction and diffusion mechanics.

Results and Discussion:

(i). Material Parameter: The simulation uses baseline transport properties for NMC cathode materials and single-crystal Silicon substrates:

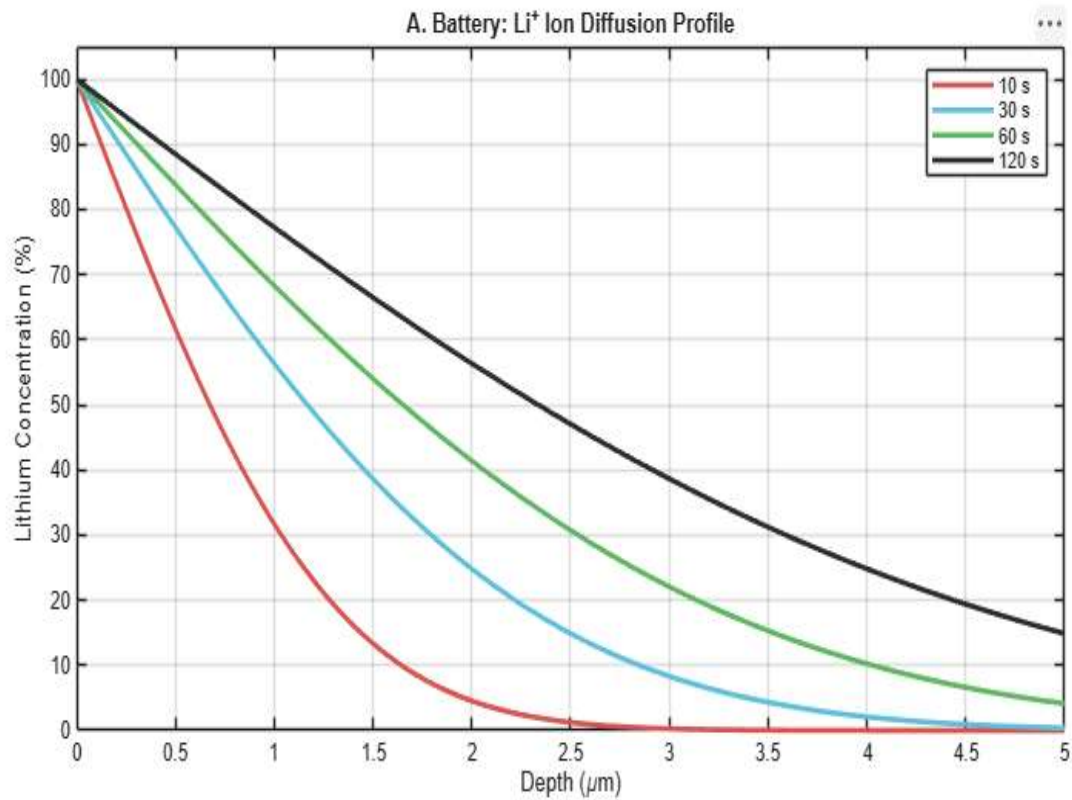
Domain	Material	Transport Parameter	Values & Units	Physical Significance
Battery (Mass)	Li ⁺ in NMC Cathodes	D_{Li}	$10^{-14} \text{ m}^2/\text{sec}$	Interstitial diffusion via open layered channels at 25°C.
Semiconductor (Mass)	Boron in Si Lattice	D_{B}	$10^{-18} \text{ m}^2/\text{sec}$	High activation barrier for substitutional lattice movement.
Battery (Thermal)	NMC in ceramic Oxide	α_{NMC}	$10^{-7} \text{ m}^2/\text{sec}$	Low thermal conductivity ($k = 1\text{-}2 \text{ W/m} \cdot \text{K}$) heat retention.
Semiconductor (Thermal)	Mono crystalline Silicon	α_{Si}	$9 \times 10^{-5} \text{ m}^2/\text{sec}$	High thermal conductivity ($k = 150 \text{ W/m} \cdot \text{K}$) rapid dissipation.

(i). Mass Transport Profile Analysis: The transient concentration profiles (C/C_0) across the 5 μm domains demonstrate a time-scale divergence between interstitial ion insertion and dopant migration.

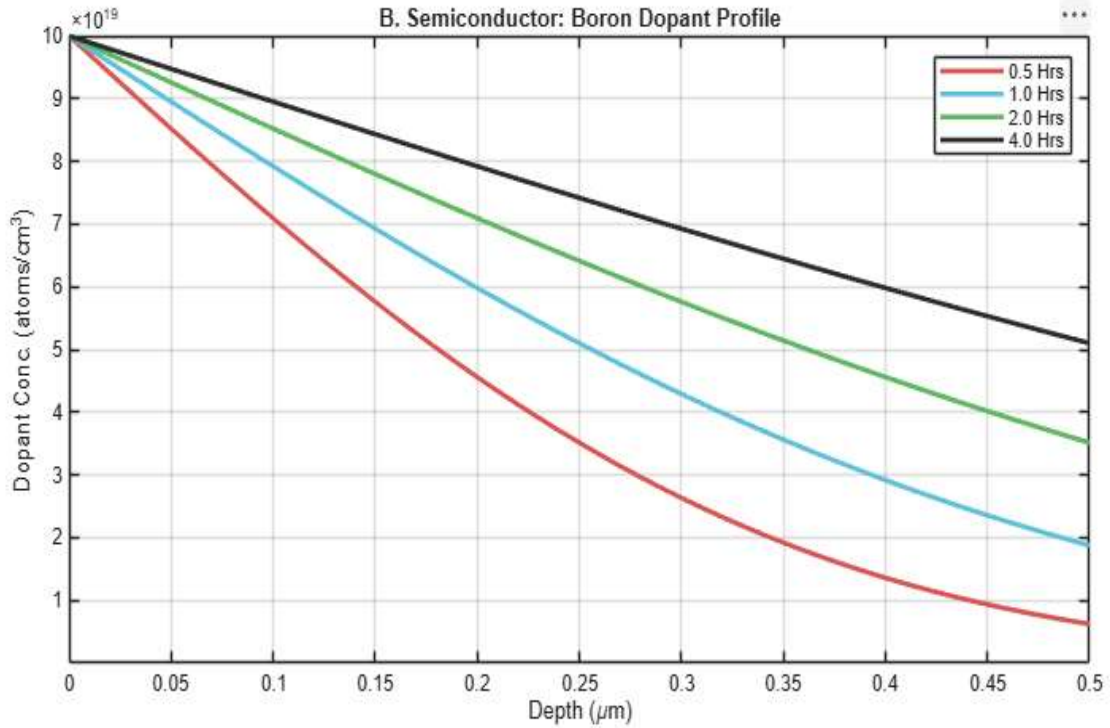
Mass penetration depth over time:

Time (t)	Li^+ in NMC cathodes ($D = 10^{-14} \text{ m}^2/\text{sec}$)	Boron in Si Lattice ($D = 10^{-18} \text{ m}^2/\text{sec}$)
10 s	0.63 μm	0.0002 μm
100 s	2.00 μm	0.0006 μm
3600 s (1 hr)	12.0 μm	0.12 μm

Subplot A:



Subplot B:



Battery Mass Transport (Subplot A): Lithium ions penetrate deep into the cathode substrate within seconds, making mass transport sufficiently fast for dynamic charging cycles.

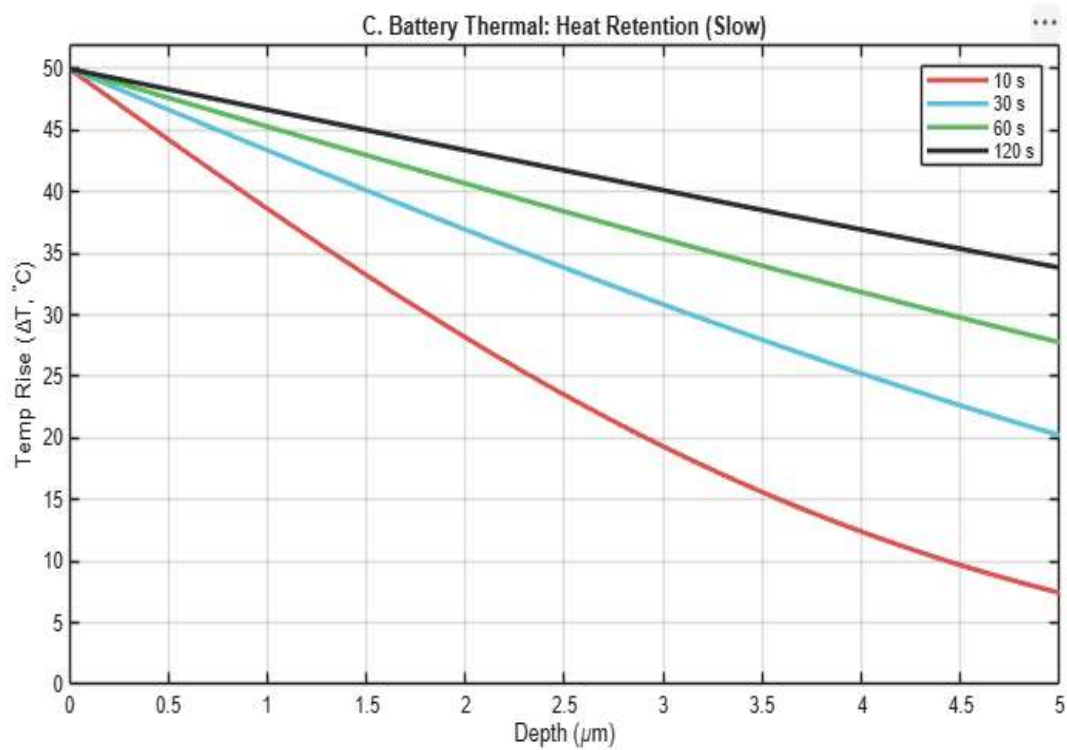
Semiconductor Mass Transport (Subplot B): Boron dopants require several hours of thermal activation (1000°C) to penetrate fractions of a micrometer ($< 0.5 \mu\text{m}$), enabling ultra-shallow, precise PN junction fabrication.

(ii). Thermal Conduction Profile Analysis: Evaluating transient heat propagation ($\Delta T/\Delta T_0$) reveals an inverse relationship relative to mass diffusion; semiconductor substrates dissipate energy nearly instantaneous compared to ceramic oxides.

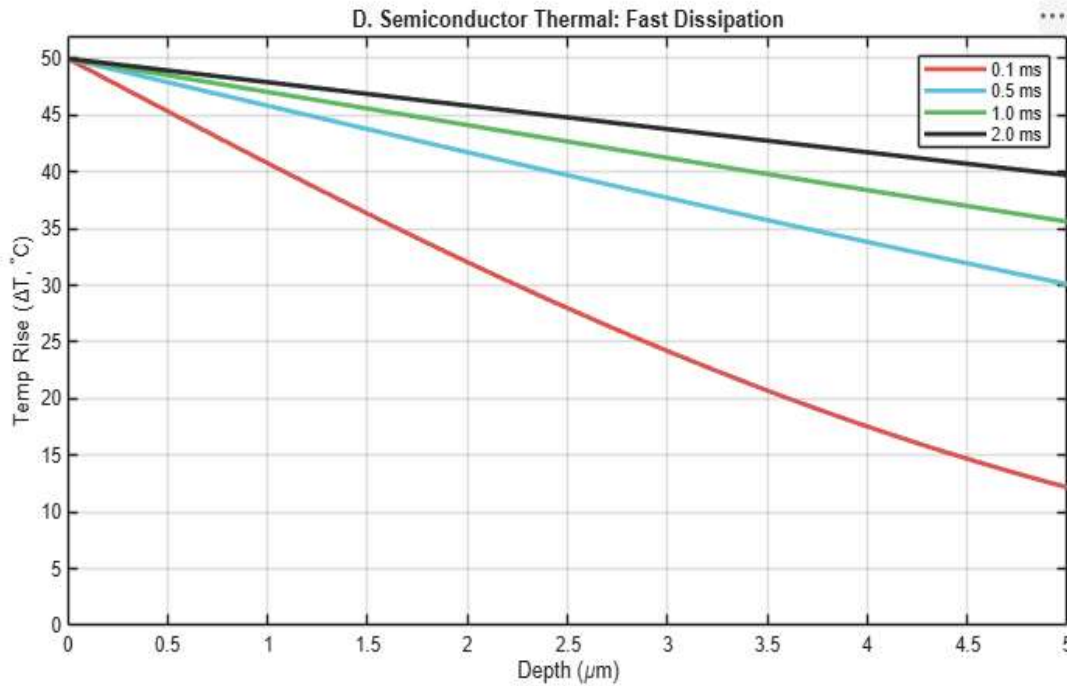
Thermal Propagation Profiles across 5 μm Substrate Depth:

Time Step (t)	NMC Cathode Temp Rise (ΔT) ($\alpha = 10^{-7} \text{ m}^2/\text{s}$)	Silicon Temp Rise (ΔT) ($\alpha = 9 \times 10^{-5} \text{ m}^2/\text{s}$)
1 μs	Localized to $x < 0.6 \mu\text{m}$	Uniform ($x > 5 \mu\text{m}$)
10 μs	X= 2.0 μm reached	Complete steady state
100 μs	Bulk equilibrium reached	Instantaneous dissipation

Subplot C:



Subplot D:



NMC Heat Retention (Subplot C): Due to strong phonon scattering mechanisms in complex ceramic oxides, NMC exhibits a low thermal diffusivity ($\alpha_{\text{NMC}} = 10^{-7} \text{ m}^2/\text{sec}$). Thermal spikes remain localized near the surface during transient pulses ($t < 10 \text{ } \mu\text{s}$), creating steep internal temperature gradients (T / x). That necessitates active liquid cooling.

Silicon Rapid Dissipation (Subplot D): Single-crystal silicon's high thermal conductivity allows thermal waves to traverse $5 \text{ } \mu\text{m}$ domains in under $1 \text{ } \mu\text{s}$ ($\alpha_{\text{Si}} = 900 \times \alpha_{\text{NMC}}$). This rapid equilibration prevents localized hotspots during high-frequency microchip switching.

MATLAB Script:

```
clear; clc; close all;

% Initialize Figure Window
figure('Name', 'Transport Dynamics Fixed', 'Position', [100, 100, 1100, 800], 'Color', 'w');

% --- Grid Setup ---
x_full = linspace(0, 5, 200);    % 0 to 5 um for Thermal & Battery
x_semi = linspace(0, 0.5, 200);  % 0 to 0.5 um Dopant Junction

% --- Battery Mass Transport (NMC Solid-State Li+ Diffusion) ---
C0_bat = 100;
D_bat_eff = 5e-2;                % Effective Li+ diffusivity in um^2/s
times_bat = [10, 30, 60, 120];  % seconds

% --- Semiconductor Mass Transport (Boron in Si at 1000°C) ---
C0_semi = 1e20;
D_semi_eff = 2e-5;               % Effective Boron diffusivity in um^2/s
times_semi_hrs = [0.5, 1, 2, 4]; % hours

% --- Thermal Parameters ---
T0_shock = 50;
alpha_bat_um = 0.6;              % NMC Thermal Diffusivity (um^2/s)
alpha_semi_um = 92000;           % Silicon Thermal Diffusivity (um^2/s)

thermal_times_bat = [10, 30, 60, 120]; % seconds
thermal_times_semi = [1e-4, 5e-4, 1e-3, 2e-3]; % seconds

colors = {'#d9534f', '#5bc0de', '#5cb85c', '#292b2c'};

% Plotting Subplots

% --- Subplot A: Battery Li+ Ion Diffusion ---
subplot(2, 2, 1); hold on; box on; grid on;
for i = 1:length(times_bat)
    t = times_bat(i);
    C_bat = C0_bat * erfc(x_full / (2 * sqrt(D_bat_eff * t)));
    plot(x_full, C_bat, 'Color', colors{i}, 'LineWidth', 2, ...
        'DisplayName', sprintf('%d s', t));
end
title('A. Battery: Li+ Ion Diffusion Profile', 'FontSize', 11);
xlabel('Depth (\um)'); ylabel('Lithium Concentration (%)');
ylim([0 105]); xlim([0 5]);
legend('Location', 'northeast'); set(gca, 'FontSize', 9);
```



```

% --- Subplot B: Semiconductor Boron Dopant Profile (FIXED: semilogy + zoomed x)|
subplot(2, 2, 2); hold on; box on; grid on;
for i = 1:length(times_semi_hrs)
    t_sec = times_semi_hrs(i) * 3600;
    C_sm = C0_semi * erfc(x_semi / (2 * sqrt(D_semi_eff * t_sec)));
    semilogy(x_semi, C_sm, 'Color', colors{i}, 'LineWidth', 2, ...
        'DisplayName', sprintf('%.1f Hrs', times_semi_hrs(i)));
end
title('B. Semiconductor: Boron Dopant Profile', 'FontSize', 11);
xlabel('Depth (\mu m)'); ylabel('Dopant Conc. (atoms/cm^3)');
ylim([1e15 1e20]); xlim([0 0.5]); % Zoomed to micro-junction depth
legend('Location', 'northeast'); set(gca, 'FontSize', 9);

% --- Subplot C: Battery Thermal ---
subplot(2, 2, 3); hold on; box on; grid on;
for i = 1:length(thermal_times_bat)
    t = thermal_times_bat(i);
    T_bat = T0_shock * erfc(x_full / (2 * sqrt(alpha_bat_um * t)));
    plot(x_full, T_bat, 'Color', colors{i}, 'LineWidth', 2, ...
        'DisplayName', sprintf('%d s', t));
end
title('C. Battery Thermal: Heat Retention (Slow)', 'FontSize', 11);
xlabel('Depth (\mu m)'); ylabel('Temp Rise (\Delta T, ^\circ C)');
ylim([0 52]); xlim([0 5]);
legend('Location', 'northeast'); set(gca, 'FontSize', 9);

% --- Subplot D: Semiconductor Thermal ---
subplot(2, 2, 4); hold on; box on; grid on;
for i = 1:length(thermal_times_semi)
    t = thermal_times_semi(i);
    T_semi = T0_shock * erfc(x_full / (2 * sqrt(alpha_semi_um * t)));
    plot(x_full, T_semi, 'Color', colors{i}, 'LineWidth', 2, ...
        'DisplayName', sprintf('%.1f ms', t * 1e3));
end
title('D. Semiconductor Thermal: Fast Dissipation', 'FontSize', 11);
xlabel('Depth (\mu m)'); ylabel('Temp Rise (\Delta T, ^\circ C)');
ylim([0 52]); xlim([0 5]);
legend('Location', 'northeast'); set(gca, 'FontSize', 9);

```

Conclusion:

This project successfully simulated non-steady-state mass and heat transport in NMC battery cathodes and Silicon semiconductors using Fick's and Fourier's transfer equations in MATLAB. The results highlight an inverse transport dynamic: Li-ion mass transport in NMC occurs rapidly at room temperature (**0-5 μm in 120 sec**), but its low thermal diffusivity ($\alpha = 0.6 \mu\text{m}^2/\text{sec}$) leads to significant heat retention. Conversely, Boron dopant diffusion in Silicon requires extreme temperatures (1000°C) over hours for shallow penetration ($< 0.5 \mu\text{m}$), yet its ultra-high thermal diffusivity enables dissipation within milliseconds ($< 2 \mu\text{s}$). Consequently, active thermal management is imperative for battery cathodes, whereas microelectronic fabrication demands precise thermal budget control during dopant drive-in.

Design Takeaway: While Silicon requires high thermal energy for mass processing but self-dissipates heat rapidly during operation, NMC cathodes process mass quickly during charging but require active liquid cooling to mitigate severe internal thermal gradients.

References:

1. **Bird, R. B., Stewart, W. E., & Lightfoot, E. N.** (2002). *Transport Phenomena* (2nd ed.). John Wiley & Sons. (*Foundational reference for non-steady-state Fickian and Fourier transport PDEs and error function solutions.*)
2. **Incropera, F. P., DeWitt, D. P., Bergman, T. L., & Lavine, A. S.** (2007). *Fundamentals of Heat and Mass Transfer* (6th ed.). John Wiley & Sons. (*Standard values for thermophysical properties, thermal diffusivity calculations, and transient heat conduction models.*)
3. **Plummer, J. D., Deal, M. D., & Griffin, P. B.** (2000). *Silicon VLSI Technology: Fundamentals, Practice, and Modeling*. Prentice Hall. (*Diffusion coefficients for substitutional boron dopants in silicon and thermal oxidation kinetic models*)
4. **Verma, P., Maire, P., & Novák, P.** (2010). *A review of the features and analyses of the solid electrolyte interphase in lithium-ion batteries*. *Electrochimica Acta*, 55(22), 6332-6341. (*Solid-state lithium-ion diffusion dynamics within NMC and graphite host matrices*)
5. **Doyle, M., Fuller, T. F., & Newman, J.** (1993). *Modeling of galvanostatic charge and discharge of the lithium/polymer/insertion cell*. *Journal of the Electrochemical Society*, 140(6), 1526-1533. (*Concentration-dependent species of transport and electrochemical kinetic boundaries*).

